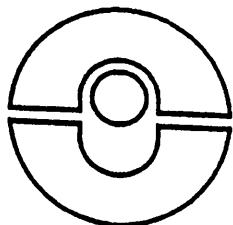


INTERNATIONAL



UNION OF RAILWAYS

U I C CODE

Leaflet to be classified in Volumes :

VII - WAY AND WORKS

VIII - TECHNICAL SPECIFICATIONS

864 -1



3rd Edition, 1-1-82

**TECHNICAL SPECIFICATION
FOR THE SUPPLY OF SLEEPER SCREWS**

(Reprint, 1-4-90)

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0 - INTRODUCTION

The present specification sets out the conditions covering the quality of the materials and the manufacture, as well as the acceptance conditions for sleeper screws used in the permanent way.

Pending the publication of international standards, this specification must be accompanied, when requesting tenders, by the documents necessary for the execution of the order and, in particular, by those concerning the application of the points in the present specification set out below :

- 1.1. - Materials,
- 1.2. - Manufacturing processes,
- 1.3. - Drawings and gauges
- 1.4. - Marks,
- 1.6. - Tolerances,
- 2.1.2.1. - Tests on materials,
- 2.1.2.2. - Tests and inspections carried out on finished parts,
- 2.3.2.2.2.- Dimensional inspection and inspection of fine workmanship,
- 2.3.2.2.3. - Derogation from qualitative and dimensional inspections by sampling from batches. - Quality control cards,
- 2.3.2.2.4. - Galvanisation inspection,
- 3.1. - Protection,
- 3.2. - Packing.

1 - CONDITIONS OF MANUFACTURE

1.1. - Materials

The steel making process shall be left to the discretion of the supplier. At the Purchasing Railway's request, the supplier must reveal the process used.

The bar steel used for the manufacture of sleeper screws is set out in the table below, according to the choice of the purchasing Railway.

	Non-hardened sleeper screws				Hardened sleeper screws
Strength category (1)	4.6	4.8	5.6	5.8	5.6
Tensile strength δB in N/mm ²	nom.		400		500
	min.		400	420	500
Elongation at fracture $\delta 5$ in %	min.		22	14	20
Limit of elasticity δS in N/mm ²	nom.		240	320	300
	min.		240	340	300
Nominal limit of elasticity $\delta_s \cdot 100$		60	80	60	80
Nominal tensile strength δB in %					

(1) The designation of the strength category includes two figures, separated by a dot

- the first figure gives 1/100th of the nominal tensile strength in N/mm².
- the second figure gives the decuple of the ratio of the nominal limit of elasticity to the nominal tensile strength.
- the multiplication of both figures gives 1/10th of the nominal limit of elasticity in N/mm².

Where national norms have to be observed, the nearest grade of steel to that selected from the above table should be chosen.

1.2 - Manufacturing processes

1.2.1 - General

Sleeper screws shall be manufactured from a single piece, without welding. Their head shall be formed by hot-working from the bar-stock.

Sleeper screws shall be hot threaded, according to the information given on the drawing by the purchasing Railway.

The request for tenders shall specify whether the sleeper screws are required to be hardened (see article 1.2.2.) and whether they are to be coated for protection against corrosion. In the latter case, the kind of protective coating required shall be indicated, together with the procedures to be adopted for ensuring its correct application ; in cases where protection is to take the form of galvanisation, see Points 1.2.3. and 2.3.4.

1.2.2 - Hardening

Sleeper screws to be hardened shall be brought to a temperature of 850°C and shall be quenched in water at a temperature of 30°C to 75°C until completely cold. Sleeper screws which are required to be hot coated may, however, be withdrawn from water before complete cooling takes place, provided that their temperature is not above 100°C.

1.2.3 - Galvanisation

Galvanisation shall be carried out in a bath of molten zinc, or by means of any other process accepted by the purchasing Railway. The zinc used shall conform to the national standards of the manufacturing country.

Sleeper screws must be free from any trace of grease, paint, etc. before the zinc coating is applied.

1.3 - Drawings and gauges

The purchasing Railway shall provide the supplier with a copy of the drawings at the same time as the advice approving the order.

The supplier shall be bound to submit two sets of gauges, both minimum and maximum, male and female, to the approval of the purchasing Railway, before manufacture begins. Provided they are correct, the gauges in question shall be stamped and one set returned to the supplier, who can make use of it for checking the gauges required for manufacture. The second set of gauges shall remain at the disposal of the receiving inspectors. Only gauges stamped by the purchasing Railway shall be valid for acceptance purposes.

The order shall also specify whether the supplier is required to send samples of each type of sleeper screw ordered (up to a maximum of three) at the same time as the gauges. If so, manufacture shall not begin until the samples have been approved.

The preparation of the gauges and the supply and despatch of the samples shall be the responsibility of the supplier. The preparation of gauges shall not be required for parts ordered in batches of less than 10 000.

1.4 - Marks

The following marks shall be shown on the upper surface of the head of sleeper screws in permanently legible embossed characters formed during manufacture :

- the supplier's mark,
- the last two figures of the year of delivery,
- the identification mark.

The drawing shall also indicate whether the top of the sleeper screw is required to bear any special letter or mark.

1.5 - Finish

Sleeper screws shall be manufactured in accordance with the drawings. In particular, care shall be taken to see that the heads are correctly centred in relation to the shanks. No folding of the metal shall occur on the heads, especially at the point of junction with the shank.

The complete surface of the screws shall be suitably trimmed. Care shall be taken to ensure that trimming does not give rise to any tearing of the metal or leave any roughness on the contact surfaces.

The threads must be clean, uniform and complete.

If the threads of sleeper screws are stripped or if there is a lack of material in their shank, they shall not be accepted.

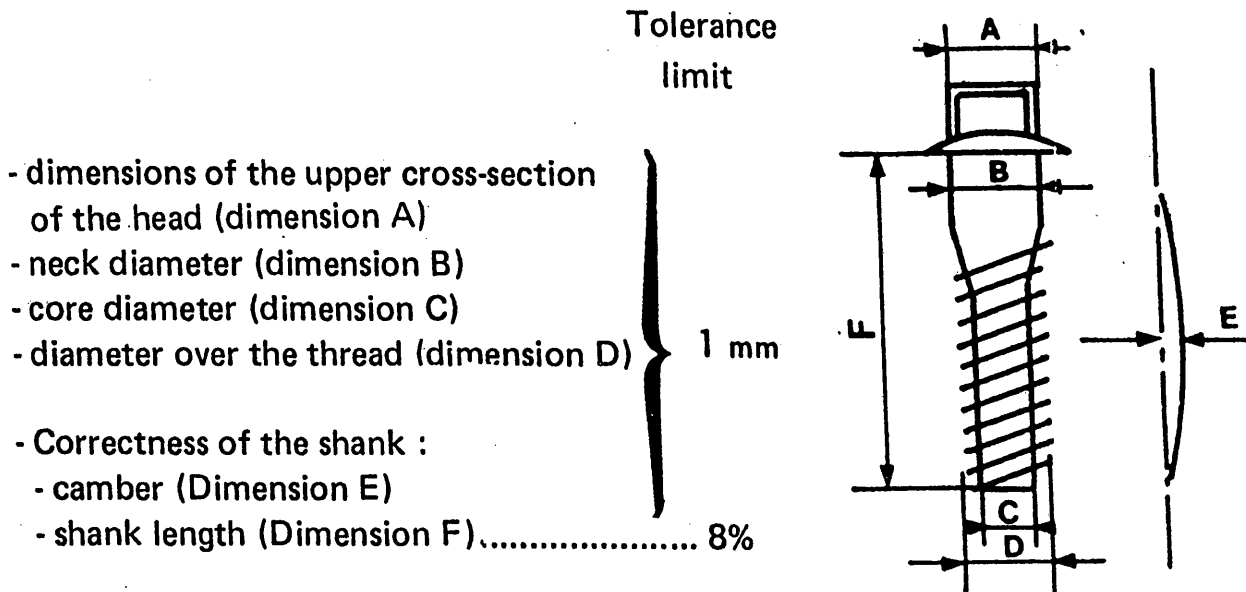
1.6 - Tolerances

1.6.1 - *Dimensional tolerances*

The tolerances to be observed in manufacture shall be shown on the drawings attached to the request for tenders.

As regards the dimensions subject to systematic dimensional inspection, as defined in Point 2.1.2.2, these tolerances shall be such that the following

tolerance limits apply :



N.B. The tolerance limit for Dimension B may be increased to 1.4 mm in the case of sleeper screws with a shank length of more than 40 mm.

The sleeper screw centre-line must not be more than 1mm distant from the shank centre-line in all radial directions.

1.6.2 - Weight tolerance

The request for tenders shall specify whether payment is to be made by weight or per part.

In the case of the former, provided the dimensional tolerances are respected, payment shall be based on the actual weight of the parts. Payment for any excess weight shall only be made up to 2% of the normal weight shown on the drawings ; in the event of a deficiency in weight occurring, the actual weight shall in all cases be the basis for payment.

The actual weight of a sleeper screw shall be considered as the average weight of 100 parts, half of which are selected by the receiving inspector, and half by the factory, for each batch submitted for acceptance. The calculation of the standard weight shall be based on the drawings, adopting a metal density of 7.85.

2 - ACCEPTANCE CONDITIONS

2.1 - General

2.1.1 - *Preliminary observations*

The steel producing factories and the supplier must be in a position to carry out all the prescribed tests.

The preparation of test pieces and tests shall be the responsibility of the supplier, who shall place the necessary staff and equipment at the receiving inspector's disposal. The parts or materials submitted for testing or inspection shall not be debited to the purchasing Railway, except in the case of parts used for the dimensional inspection (Point 2.3.2.2.2) and found to be satisfactory.

2.1.2 - *Nature and proportion of tests*

2.1.2.1 - *Tests on materials*

Acceptance of the steel bars shall take place in accordance with the instructions contained in the request for tenders, either on the premises of the suppliers of sleeper screws, or in the producing factories. These bars shall be subjected to the following mechanical tests :

- | | | |
|----------------|---|---------------------------------------|
| - tensile test | } | one series of tests per weight of 10t |
| - bend test | | |

It is necessary to specify, in the request for tenders, whether the purchasing Railway accepts a rolling mill certificate for each cast (20 t).

2.1.2.2 - Tests and inspections carried out on finished parts

Sleeper screws shall be subjected to the following tests and inspections :

- mechanical tests
 - tensile test
 - bend test
- } one series of tests per batch of 2000 to 100 000 parts
(see Points 2.2.2 and 2.3.2.2.1)
- dimensional inspection and finish inspection : one series of inspections per batch (see Points 2.2.2 and 2.3.2.2.2).

The dimensional inspection shall be carried out on the dimensions marked A to F. The other dimensions shown in the standards and drawings accompanying the order may be checked at any time by the receiving inspector, but are not subjected to the systematic inspection referred to in Point 2.3.2.2.2.

- Inspection of galvanisation (if any) : one series of tests per batch of 2000 to 100 000 parts (see Point 2.2.2 and 2.3.2.2.4).

The purchasing Railway can derogate from the tests and inspections in accordance with the conditions laid down in Point 2.3.2.2.3.

The mechanical tests, the dimensional inspection and finish inspection shall normally be carried out in the producing factory, before any zinc coating is applied. The galvanisation inspection shall be made at the workshop where the galvanising operation was carried out.

Note : In cases where the sleeper screws are not required to be hardened, the tensile test on finished parts can be dispensed with if the material has been tested.

2.2 - Sampling and preparation of test pieces, sampling of finished parts

2.2.1 - Tests on materials

The bars selected for testing shall be cut up and marked in accordance with the instructions of the receiving inspector.

Cutting and machining shall take place entirely when cold, by means of machine tools, without having recourse to any hammering, cold bending, hardening or annealing.

Test pieces shall retain the inspector's mark intact.

The gauge length for calculating the elongation shall be that recommended by the ISO i.e. $L_0 = 5.65 \cdot \sqrt{S_0}$ for test pieces with a prismatic cross-section (S_0 being the original cross-section of the calibrated part), and $L_0 = 5d_0$ for test pieces with a circular cross-section (d_0 being the diameter).

2.2.2 - Tests and inspections on finished parts

The selected parts shall be grouped together in batches of the same kind. Samples shall be taken in such a way that they are representative of the batches submitted. A batch shall contain not less than 2000 or more than 100 000 parts.

The receiving inspector shall have the right to break down or form batches for the purpose of testing and inspection.

Finished parts intended for testing shall be marked by the receiving inspector and must retain these marks intact until acceptance has been completed.

2.3 - Carrying out of the tests and inspections - Results to be obtained

2.3.1 - General conditions

The limits laid down for the results to be obtained from the tests or inspections must be fully observed in respect of each part.

In order to be accepted, the batch of parts submitted must satisfy all the prescribed tests and inspections.

2.3.2 - Procedure for tests and inspections

2.3.2.1 - Tests on materials

If a particular part chosen from a batch for a certain test does not satisfy the conditions laid down, two similar check tests may be carried out at the request of the supplier. If one of these check tests does not prove to be satisfactory, the batch concerned shall be rejected.

2.3.2.2 - Tests on finished parts

2.3.2.2.1 - Mechanical tests

The mechanical tests shall be carried out by taking samples from batches of sleeper screws.

They shall be carried out in accordance with the progressive Wald method (see Appendices A and B) or any other method for statistical quality inspection approved by the purchasing Railway. The purchasing Railway shall attach to the request for tenders the model diagram to be used.

2.3.2.2.2 - Dimensional inspection and inspection of fine workmanship

Dimensional inspection and inspection of fine workmanship shall be effected by means of sampling from batches of sleeper screws.

They shall be carried out in accordance with the progressive Wald method (Appendix 1) or any other method for statistical quality inspection approved by the purchasing Railway. The purchasing Railway shall attach to the request for tenders the model diagram to be used.

In the event of rejection, the supplier shall have the option of sorting the parts of the faulty batch, at his own expense, and to re-submit the batch for acceptance.

For the second submission, the inspection shall be carried out in accordance with a progressive plan (Appendix 2) which provides a smaller risk for the purchasing Railway.

2.3.2.2.3 - Derogation from qualitative and dimensional inspections by sampling from batches. Quality control cards

In cases where the supplier makes a regular practice of using a system of quality control cards agreed by the purchasing Railway for his products, the Railway can arrange the frequency of the dimensional and qualitative inspections by sampling from batches, as it thinks best. The results recorded on the quality control cards shall then be considered as constituting an acceptance inspection.

The agreement of the Railway concerning the use of quality control cards shall cover, in particular, the following points :

- the manufacturing stages at which the control is carried out,
- the extent and frequency of sampling,
- the limits of the control,
- where applicable, the number of parts outside the limits, allowed during a certain period.

The quality control cards shall be placed at the receiving inspector's disposal who shall be free, moreover, at any time, to inspect the correct application of the procedure by any method chosen by the Railway.

The quality control cards must contain any information necessary for the purpose of clearly identifying the product. They shall be retained by the supplier at least until 31st December of the year following the year shown on the parts.

2.3.2.2.4 - Galvanisation inspection

Galvanisation inspection shall be carried out by taking samples from batches of sleeper screws.

It shall be effected in accordance with the progressive Wald method (Appendices A and B) or any other method for statistical quality control approved by the purchasing Railway. The purchasing Railway shall attach to the request for tenders the model diagram to be used.

2.3.3 - *Carrying out of tests and inspections*

2.3.3.1 - *Tests on material*

These tests shall be carried out in accordance with the national standards of the country of the supplier.

2.3.3.1.1 - Tensile test

For the purpose of this test, the steel for testing shall be of the required quality, based on one of the grades indicated in Point 1.1.

2.3.3.1.2 - Bend test

During this test no cracks, scaling or other fault whatsoever shall occur.

2.3.3.2 - Tests on finished parts**2.3.3.2.1 - Tensile test**

The sleeper screw shall be secured by both ends in the grips of a tensile machine and stretched until fracture occurs. The grips shall be specially prepared for the purpose of this test and shall ensure that the load is centrally placed in relation to the part to be tested.

On no account shall fracture occur at the point where the shank of the sleeper screw joins the head.

The breaking load per mm^2 of the original cross-section of the shank or of the cross-section measured at the bottom of the thread (depending on whether fracture occurred in the smooth portion or in the threaded portion), shall not be less than the tensile strength laid down for the material.

In the case of hardened sleeper screws, the tensile strength must be between 670 and 920 N/mm^2 .

2.3.3.2.2 - Bend test

The shank shall be evenly bent in the threaded part, on a mandrel with a diameter equal to $4d$, to an angle of :

- 90° in the case of sleeper screws in non-hardened steel,
- 15° in the case of sleeper screws in hardened steel.

During this test, no cracks, scaling or other fault whatsoever shall occur.

2.3.3.3 - *Dimensional inspection and inspection of fine workmanship*

Any sleeper screw with at least one inspected dimension outside the tolerances or which does not satisfy the conditions relating to finish given in Point 1.5, shall be considered as «bad» for the purpose of the inspection.

2.3.4 - *Galvanisation inspection*

Galvanised parts may possess either a matt or a gloss finish. Galvanised surfaces shall be even, smooth, and possess no swelling, drips, cracks, grittiness or other fault whatsoever.

The thickness of the coating shall be checked in accordance with the conditions prescribed by the purchasing Railway. In the case of an electromagnetic method, the coating thickness shall be measured on the head and on the shank. It must comply with the particulars given in the drawing or in the special document of the purchasing Railway.

The check shall be carried out by sampling from batches of sleeper screws (see Point 2.3.2.2).

3 - DELIVERY STATE OF THE SUPPLIES

3 - Protection

Sleeper screws not provided with a durable protective coating must be protected against rust during long periods of storage. The request for tenders shall indicate the method of protection required by the purchasing Railway.

Irrespective of the method of protection adopted, all surfaces must be thoroughly covered with a rustproof coating.

3.2 - Packing

The order shall indicate the kind of packing required for each supply (hampers, barrels, boxes, metal casks or sacks), together with the number of parts per package or the maximum weight for each package.

Packages used for despatch shall not have been used previously for any purpose which might cause it to damage the material.

The order shall also state whether the packages must be sealed by the receiving inspector with the purchasing Railway's stamp. In this event, the supplier shall place the necessary tools and material for sealing at the disposal of the receiving inspector without additional charge, and give him any assistance required.

Each package shall bear the following information as applicable, in clear indelible characters on a label firmly fixed to the package :

- the name or the mark of the supplier,
- the order number,
- the description of the parts,
- the number and total weight of the parts.

Unless the packing is of the type specially designed for repeated use, it shall not be charged for or returned to the supplier and shall remain the property of the purchasing Railway. Where the contrary is the case, the necessary instructions shall be shown in the request for tenders.

4 - GUARANTEE (1)

The supplier shall guarantee his deliveries for a period extending up to 31st December of the year following the year shown on the parts.

(1) The guarantee clause does not apply to British Railways.

Neither the supervision by the persons delegated by the purchasing Railway nor the tests carried out at the time of acceptance shall diminish in any way the liability of the supplier.

Sleeper screws, which, during the guarantee period, show defects making them either unfit for service or reducing their period of service, shall be made available to the supplier with a view to their replacement or reimbursement.

NOTE ON THE USE OF THE WALD DIAGRAM

When a test is carried out, samples are selected at random, one after the other, from the batch being tested. Each of them is tested and the result is recorded on a diagram before the next one is selected.

A sample shall be deemed «good» if the test or check referred to in the diagram is satisfactory. It shall be deemed «bad» if the contrary is the case.

The results appear in the diagram as a point moving across the field. The starting position of the point is at the origin of the coordinates. Every time a test is performed, the point is replotted forward by one unit parallel to the x-axis ; for each «bad» test, it is also replotted upward by one unit parallel to the y-axis. The test is stopped as soon as the recording point enters one of the areas marked «accept» or «reject».

Various kinds of tests, making up one series, may be plotted on the same diagram.

The diagrams relating to each series of tests shall be appended to the acceptance report. They must show every consecutive position of the recording point.

The purchasing Railway may allow the sampling to be carried out by takings, instead of by successive samples, operating as follows :

- 1) the number of samples composing a taking shall be the same throughout the test and be determined beforehand with a maximum number of 10 ;
 - 2) the position of the recording point shall be plotted on the diagram after all the samples of the taking have been tested ; the recording point is replotted forward after each taking, parallel to the x-axis, by as many units as there are samples in the taking, and upward, parallel to the y-axis, by a number of units equal to the number of «bad» specimens found in the taking.
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OBSERVATIONS REGARDING APPENDICES A AND B

The risks incurred in the plan represented by the diagrams (Appendices A and B) are as follows :

- For batches of 2000 to 20 000 parts :
 - a 10% maximum probability of rejecting a batch containing no more than 20% of faulty parts,
 - a 10% maximum probability of accepting a batch containing no less than 30% of faulty parts.
- For batches of 20 000 to 100 000 parts :
 - a 10% maximum probability of rejecting a batch containing no more than 10% of faulty parts,
 - a 10% maximum probability of accepting a batch containing no less than 20% of faulty parts.

The test shall be terminated as soon as the point representing the progress of the check enters the acceptance or rejection area.

OBSERVATIONS REGARDING APPENDICES 1 AND 2

The risks incurred in the plan represented by the diagram (Appendix 1) are as follows :

- a 5% maximum probability of rejecting a batch containing no more than 5% of faulty parts,
- a 5% maximum probability of accepting a batch containing no less than 15% of faulty parts.

The check shall be terminated as soon as the point representing the progress of the test enters the acceptance or rejection area.

In the event of rejection, the supplier shall have the option of sorting the parts of the faulty batch, at his own expense, and of submitting the batch again for acceptance.

For the second submission, the inspection shall be carried out in accordance with a progressive plan (Appendix 2) which provides a smaller risk for the purchasing Railway.

RAILWAY :

Department :

Acceptance of
SLEEPER SCREWS

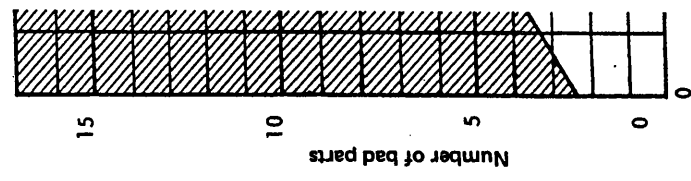
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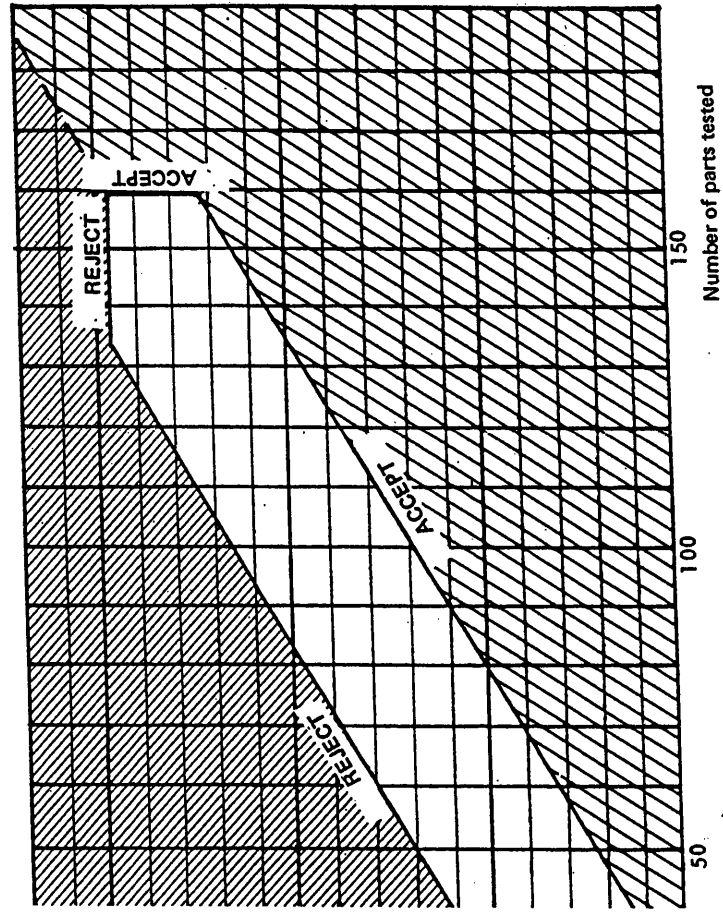
Dimensional and qualitative inspection

OBSERVATIONS :

Suppli



WALD DIAGRAM



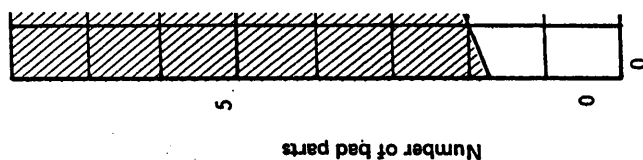
RAILWAY :

Department :

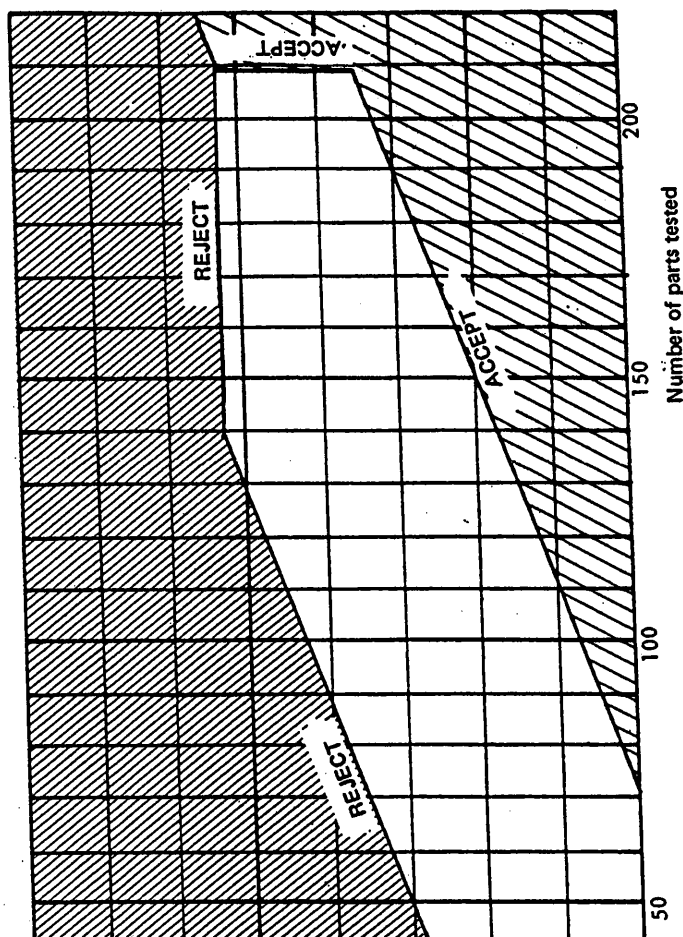
Acceptance of
SLEEPER SCREWS
Type :
Date :
Dimensional and
qualitative inspection
(after sorting)

OBSERVATIONS :

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WALDDIAGRAM

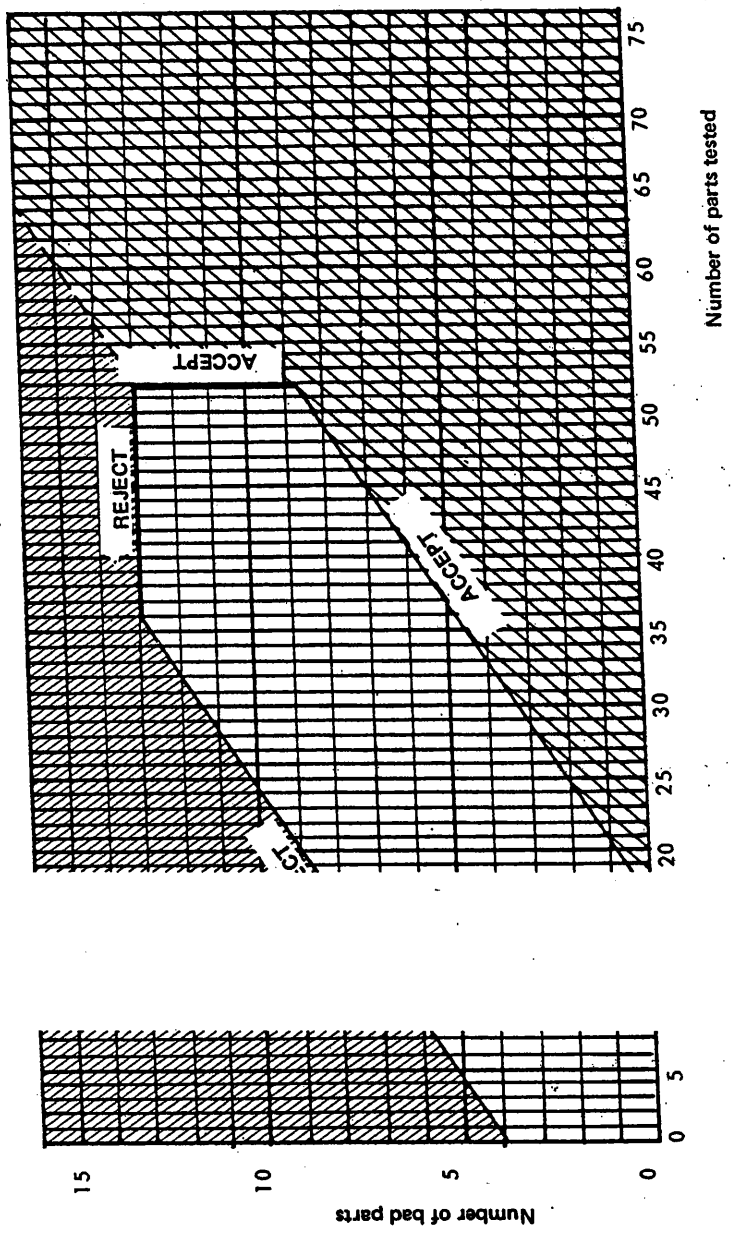


RAILWAY

Department

Acceptance of
SLEEPER SCREWS
Type :
Date :
Mechanical tests :
- Tensile strength test
- Bend test
Galvanisation inspection

Supplier



OBSERVATIONS

RAILWAY

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Department :

Acceptance of
SLEEPER SCREWS

Type :

Date :

Mechanical tests :

- tensile strength test

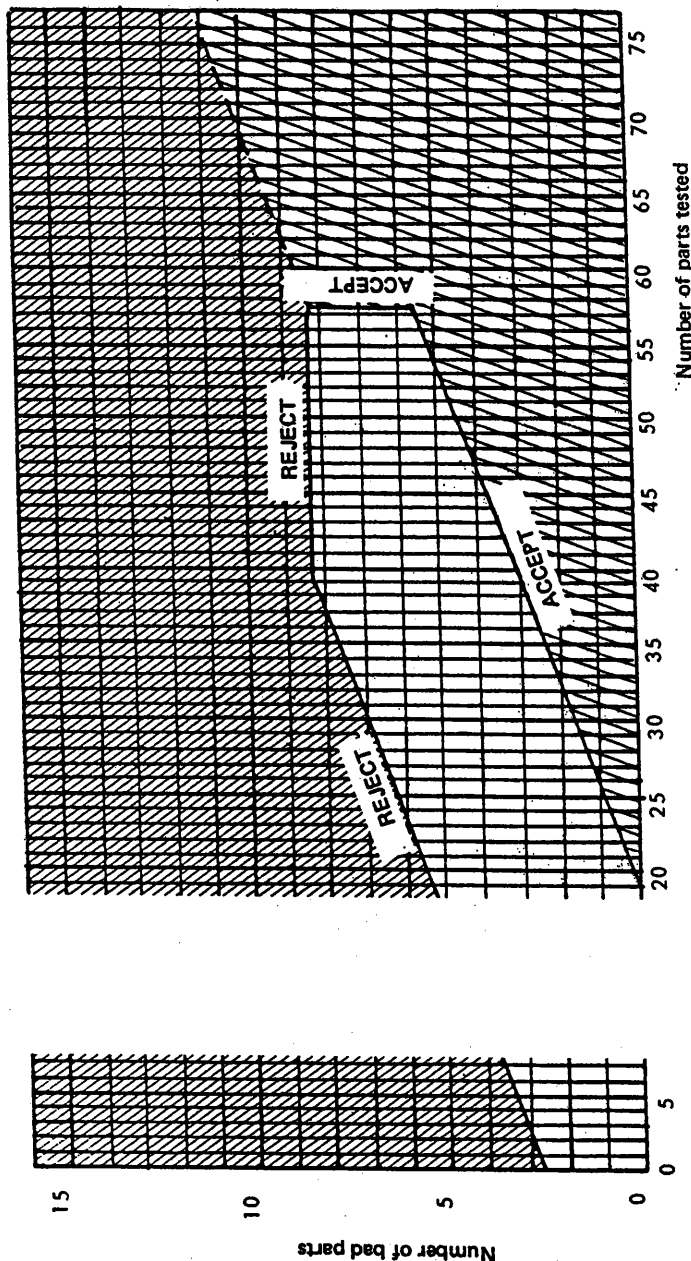
- Bend test

Galvanisation inspection

OBSERVATIONS :

WALD DIAGRAM

for batches of 20 001 to 100 000 parts



APPLICATION

With effect from 1 January 1982.

All Railways in the Union.

RECORD REFERENCES

Headings under which the question has been dealt with :

- Preparation of specifications for the supply of permanent way material :

- a) sleeper screws and fishplate bolts,
- b) metal sleepers for standard gauge track,
- c) single-unit points and crossings in manganese steel.

(5th Committee -P.W.-S- : Copenhagen, May 1956 - Board of Management : December 1957).

- Full application of statistical control for the acceptance of sleeper screws.

(5th Committee -P.W.-S- : Budapest, June 1958 - 7th Committee -P.W. : Prague, June 1959 ; Stuttgart, May 1960 ; Paris, May 1961).

- Full application of statistical control for the acceptance of sleeper screws, fishplates, sole-plates and elastic washers.

(7th Committee -P.W.- : Portsmouth, May 1962).

- Revision of Leaflet 864-1.

(Sub-Committee for Permanent Way and Maintenance : Paris, January 1979 ; Way and Works Committee : Madrid, June 1981).